ANSWER KEY & MARKING SCHEME SET 1

Chapter 3: **Shaping the Earth's Surface (Geography)** | Class 9 Social Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** transform boundary	No crust created or destroyed
2	**(c)** steep, wet and deforested	Natural and human triggers
3	**(a)** crust	Outer solid layer
4	**(b)** avalanche	Snow slide
5	**(d)** dissolving by acidic water	Karst
6	**(a)** bunds	Bunding
7	**(c)** slowly, a few centimetres a year	Slow movement
8	**(b)** cut-off meanders	River loops
9	**(b)** fjord	Glacial origin
10	**(c)** sand and pebbles	Shore deposit

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Rapid sliding of rock and soil downslope (1); Triggered by rain or earthquakes (1).	1+1
12	Breaking of rocks into smaller pieces (1); Happens in place (1).	1+1
13	Deposit at a river mouth (1); Fan or triangle shaped (1).	1+1
14	Winding bend in a river (1); Found in the middle course (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Crust - outer solid (1); Mantle - thick and hot (1); Core - iron and nickel (1).	1+1+1
16	Water and fertile soil (1); Transport (1); Fishing (1).	1+1+1
17	Contouring (1); Bunding (1); Terracing (1).	1+1+1
	OR: Mountain - Himalaya (1); Plateau - Deccan (1); Plain - Ganga plain (1).	1+1+1
18	Sand and sea (1); Recreation (1); Fishing and shell collecting (1).	1+1+1
	OR: Around the Pacific Ocean (1); Many earthquakes (1); Many volcanoes (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Loss of fertility (1); Washed-away land (1); Contouring (1); Bunding and terracing (1); Check dams (1).	1+1+1+1+1
	OR: Cliffs (1); Wave-cut platforms (1); Caves (1); Arches and stacks (1); Beaches (1).	1+1+1+1+1
20	Physical caused by temperature (1); Chemical by water and air (1); Biological by plants and animals (1); Examples (1); Soil forms (1).	1+1+1+1+1
	OR: Baratang in Andaman and Nicobar (1); Mud bubbles out due to gases (1); Bhukampa in early India (1); Varahamihira recorded signs (1); Gujarat 2001 earthquake (1).	1+1+1+1+1

Mistakes to avoid:

- **Weathering** breaks rocks in place; **erosion** carries material away by water, wind or ice. They are not the same process.
- Convergent boundary = plates move towards each other (Himalaya); divergent = move apart (mid-ocean ridges); transform = slide past (San Andreas Fault).
- Oceanic crust is thinner (about 5-7 km) than continental crust (about 30-40 km); the mantle is the thickest layer and the inner core the densest.
- Sea-coast sequence is cave, arch, stack; do not mix glacial features (cirque, fjord, U-valley, moraine, arete) with river features (V-valley, meander, oxbow lake, delta).
- A delta is a deposit at the river mouth in the lower course; an oxbow lake is a cut-off meander; waterfalls and V-shaped valleys belong to the upper course.

- Barchans and ventifacts are desert (wind) features and stalactites are limestone features; the Zabo system is from Nagaland and terraces or bunds help check erosion.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapter 3: **Shaping the Earth's Surface (Geography)** | Class 9 Social Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** glacier	Ice erosion
2	**(b)** GLOF	Outburst
3	**(c)** landforms	Natural features
4	**(c)** mantle	About 2,900 km
5	**(b)** fold mountains	Formed by collision
6	**(a)** moraine	Glacial deposition
7	**(a)** erosion	Movement
8	**(d)** meanders	Middle course
9	**(d)** wave erosion	Undercutting
10	**(d)** wind	Wind deposition

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Cutting steps on hillsides (1); Prevents soil erosion (1).	1+1
12	Mass of snow sliding down (1); Triggered by snowfall or tremors (1).	1+1
13	Mantle (1); Core (1).	1+1
14	Hill of sand (1); Formed by wind deposition (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Yardang (1); Dune (1); Desert pavement (1).	1+1+1
16	Steep gradient (1); Fast erosion (1); V-shaped valleys and waterfalls (1).	1+1+1
17	Water forms valleys (1); Wind forms dunes (1); Glacier makes U-valleys (1).	1+1+1
	OR: Slow flow (1); Heavy deposition (1); Delta and levees (1).	1+1+1
18	Cliff (1); Arch (1); Stack (1).	1+1+1
	OR: Convergent (1); Divergent (1); Transform (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Erodes sand and dust (1); Makes yardangs and ventifacts (1); Deflation hollows (1); Dunes are deposited (1); Dune types vary (1).	1+1+1+1+1
	OR: Contouring (1); Bunding (1); Terracing (1); Check dams (1); Zabo system (1).	1+1+1+1+1
20	Convergent: collision and fold mountains (1); Divergent: new crust and ridges (1); Transform: earthquakes (1); Himalaya example (1); San Andreas Fault example (1).	1+1+1+1+1
	OR: Landslide is sliding of rock and soil (1); Avalanche is a mass of snow sliding (1); Heavy rain or snowfall triggers them (1); Deforestation worsens (1); Afforestation and drainage help (1).	1+1+1+1+1

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 3: **Shaping the Earth's Surface (Geography)** | Class 9 Social Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** divergent boundary	Magma rises
2	**(c)** outer core	Iron and nickel
3	**(a)** deltas	Deposition
4	**(b)** upper course	Steep drop
5	**(c)** weathering	Surface process
6	**(b)** divergent boundary	Plates moving apart
7	**(a)** Pacific Ocean	Ring of Fire
8	**(b)** terraces	Terracing
9	**(a)** biological weathering	Plants
10	**(a)** physical weathering	Expansion

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Large slab of the lithosphere (1); Moves slowly (1).	1+1
12	A large mass of moving ice (1); Erodes valleys (1).	1+1
13	Wearing away and carrying of material (1); By water, wind, ice or waves (1).	1+1
14	Icicle-shaped hanging formation (1); In limestone caves (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Weathering in place, erosion moves material (1); Weathering has no agent (1); Erosion uses water, wind, ice (1).	1+1+1
16	Cirque (1); Arete (1); Fjord (1).	1+1+1
17	Lateral (1); Medial (1); Terminal (1).	1+1+1
	OR: Cave (1); Stalactite (1); Sinkhole (1).	1+1+1
18	Heavy rain (1); Steep slope (1); Deforestation (1).	1+1+1
	OR: Heavy snowfall (1); Weak snow layers (1); Sudden temperature rise (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Upper course: waterfalls (1); Middle: meanders (1); Lower: deltas (1); Oxbow lakes (1); Floodplains (1).	1+1+1+1+1
	OR: Glacial lake formed by melting (1); Dam of ice or moraine (1); Breaks suddenly (1); Floods downstream (1); Chamoli 2021 (1).	1+1+1+1+1
20	Crust is the thin outer layer (1); Mantle is thick and hot (1); Outer core is liquid (1); Inner core is solid and densest (1); Lithosphere and asthenosphere (1).	1+1+1+1+1
	OR: Shield from invaders (1); Passes like Khyber allowed trade (1); Rivers support farming (1); Silk Route (1); Coasts for trade (1).	1+1+1+1+1

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